

SIMATS ENGINEERING
ASSESSMENT TOOL 3
CASE STUDY BASED

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COURSE NAME: INTERNET PROGRAMMING

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Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters

multiple issues related to HTML elements, HTTP methods, XML syntax, and web

communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>

B. <video src="video.mp4" controls></video>

C. <movie src="video.mp4"></movie>

D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET

B. POST

C. PUT

D. DELETE

Snippet 3: Navigation Elements

A. <div>

B. <nav>

C. <section>

D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>

B.
<note><to>User</to><from>Admin</from></note>

C. <note><to>User</from></note>

D. <note to="User"></note>

Snippet 5: Web Communication Protocols

A. FTP

B. HTTP

C. SMTP

D. SNMP

Questions:

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B. `<video src="video.mp4" controls></video>`

Justification:

- The `<video>` tag is the standard HTML5 element used to embed videos in a web page.
- The controls attribute provides built-in play, pause, volume, and fullscreen controls for users.

Why the other options are incorrect:

- **A.** `<media>` – Not a valid HTML tag.
- **C.** `<movie>` – Not a valid HTML tag.
- **D.** `<vid>` – Not a valid HTML tag.

2. Identify the appropriate HTTP method for form submission and explain why.

Justification:

- The **POST** method is used to securely send user data (such as registration details) to the server.
- Data is sent in the **request body**, making it more suitable for sensitive information like usernames and passwords.
- There is no practical limit on the amount of data that can be sent.

Why the other options are incorrect:

- **A. GET** – Sends data in the URL, making it visible and less secure.
- **C. PUT** – Mainly used to update existing resources.
- **D. DELETE** – Used to remove resources from the server.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. `<nav>`

Justification:

- The `<nav>` element is the semantic HTML5 tag specifically designed to contain navigation links.
- It improves website accessibility and helps search engines understand the page structure.

Why the other options are incorrect:

- **A.** `<div>` – Generic container with no semantic meaning.

- **C.** <section> – Used to group related content, not specifically for navigation.
 - **D.** <header> – Represents introductory content and may contain navigation, but it is not dedicated to navigation.
4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: B.

XML

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Justification:

- All tags are properly opened and closed.
- The XML document is well-formed.

Errors in other options:

- **A.**

XML

```
<note><to>User</to><from>Admin</from>
```

Error: Missing closing </note> tag.

- **C.**

XML

```
<note><to>User</from></note>
```

Error: Mismatched tags (<to> is closed with </from>).

- **D.**

XML

```
<note to="User"></note>
```

Error: Missing closing angle bracket (>) in the closing tag, making the XML invalid.

5. Choose the correct protocol for web communication and justify your choice.

Justification:

- **HTTP** is the standard protocol used for communication between web browsers (clients) and web servers.
- It supports requests and responses for web pages, images, videos, forms, and other web resources.
- Secure communication is provided using **HTTPS** (HTTP Secure).

Why the other options are incorrect:

- **A. FTP** – Used for file transfer between systems.
- **C. SMTP** – Used for sending email.
- **D. SNMP** – Used for network monitoring and management.